

MOLECULAR BIOLOGY & GENETICS DEPARTMENT

CURRENT TOPICS IN ECOLOGY
TERM PROJECT REPORT

"Projected Climate Warming in Turkey for the European Roller (Coracias garrulus) under CMIP6 SSP Scenarios (2041–2060)"

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Abstract

The European Roller (Coracias garrulus) is a migratory bird that breeds in parts of Turkey and may be sensitive to climate-driven shifts in habitat conditions. In this project, 7,818 occurrence records from Turkey were compiled and their spatial and temporal patterns were visualized (Figure 3). Present-day climatic conditions were compared with mid-century projections (2041–2060) under two CMIP6 Shared Socioeconomic Pathways (SSP2-4.5 and SSP5-8.5) using the MIROC6 global climate model. Focusing on annual mean temperature (BIO1), future warming patterns were mapped and a difference layer (SSP5-8.5 minus current) was calculated (Figure 2). The results indicate widespread warming across Turkey by mid-century, with particularly strong increases in inland regions, suggesting potential shifts in climatic exposure relevant to the species' breeding distribution.

Introduction

The European Roller (Coracias garrulus) is a long-distance migratory bird breeding across southeastern Europe and western Asia, including large parts of Turkey. Turkey represents an important breeding and summering region for the species. Climatic factors, especially temperature, play a key role in shaping ecological processes such as prey availability, breeding phenology, and habitat suitability. Ongoing climate change is therefore expected to influence the spatial distribution of favorable climatic conditions for C. garrulus within its breeding range.



Figure 1: European Roller (*Coracias garrulus*). Photographed by Rajkumar Das (2022). Source: Macaulay Library (ML495818251).

The aim of this study was to (i) summarize and visualize the occurrence patterns of C. garrulus in Turkey using biodiversity records, and (ii) assess how annual mean temperature is projected to change by mid-century under contrasting future climate scenarios. By comparing current climatic conditions with SSP2-4.5 and SSP5-8.5 projections, this project provides a scenario-based assessment of potential climate exposure for the species in Turkey.

Materials & Methods

- **Study area:** The analysis was conducted for Turkey using national boundaries obtained from the GADM database.

- **Occurrence data:** Occurrence records for Coracias garrulus were obtained from the Global Biodiversity Information Facility (GBIF). Records with valid geographic coordinates located within Turkey were retained, resulting in a total of 7,818 observations. Temporal summaries (year and month histograms) were used to inspect sampling effort and seasonality (Figure 3).

- **Climate data:** Present-day bioclimatic conditions were obtained from the WorldClim database. Future climate projections were derived from CMIP6 using the MIROC6 global climate model for the mid-century period 2041–2060 under two Shared Socioeconomic Pathways: SSP2-4.5 (intermediate emissions) and SSP5-8.5 (high emissions).

- **Analysis and visualization:** All analyses were conducted in R using the "terra" and "geodata" packages. Annual mean temperature (BIO1) was selected as a representative climatic variable. A warming impact layer was calculated as the difference between future and current conditions (ΔBIO1 = SSP5-8.5 − current). All maps were exported as PNG figures for inclusion in the final report.

Results

Occurrence records of C. garrulus were widely distributed across Turkey, with observations concentrated during the late spring and summer months, consistent with the species' breeding season (Figure 3). The temporal distribution of records indicates that most observations occur between May and August, reflecting seasonal presence rather than year-round residency.

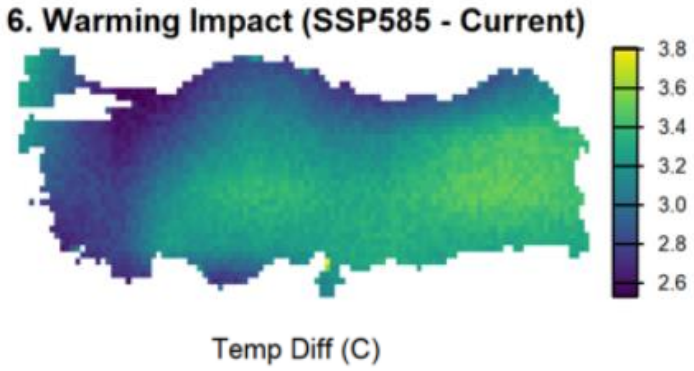


Figure 2. Occurrence records of *Coracias garrulus* in Turkey (top left), temporal distribution of records by year and month (top middle and right), projected annual mean temperature (BIO1) under SSP2-4.5 and SSP5-8.5 scenarios for 2041–2060 (bottom left and middle), and the warming impact expressed as the difference between SSP5-8.5 and current conditions (bottom right).

Climate projections show a clear warming trend across Turkey by 2041–2060 (Figure 2). Under the SSP5-8.5 scenario, annual mean temperature increases are observed throughout the country, with the strongest warming signal in inland regions, while coastal areas exhibit comparatively lower increases. The warming impact map highlights pronounced spatial heterogeneity in climate change exposure across Turkey.

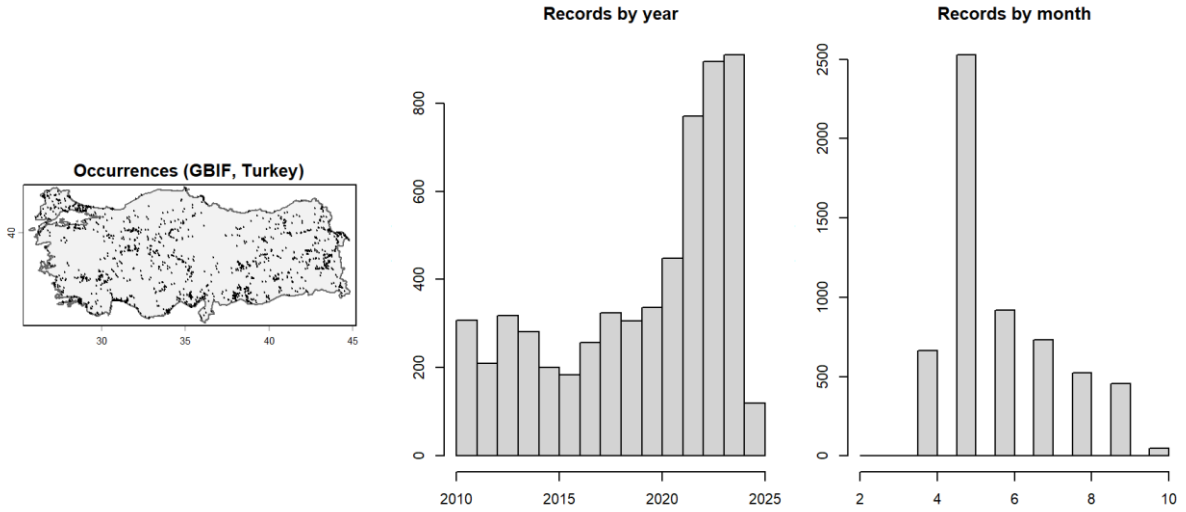


Figure 3. Spatial distribution of *Coracias garrulus* occurrence records in Turkey based on GBIF data. The left panel shows the geographic locations of observations, while the middle and right panels illustrate the temporal distribution of records by year and by month, respectively. The monthly distribution highlights a clear concentration of records during late spring and summer, corresponding to the breeding season of the species in Turkey.

Discussion

The results suggest that Turkey will experience substantial warming by mid-century, which may influence the climatic conditions encountered by C. garrulus during its breeding season. As illustrated in Figure 2, inland regions are projected to experience stronger temperature increases, potentially altering habitat suitability and thermal conditions during the breeding period. Given the species' association with warm, open habitats, moderate warming could initially expand suitable conditions in cooler or higher-elevation areas. However, stronger temperature increases in already warm inland regions may lead to thermal stress or altered ecological interactions, potentially affecting breeding success and habitat use. Moreover, this study employed a climate-based mapping and scenario-comparison approach rather than a full predictive ecological niche model with quantitative accuracy metrics such as AUC or TSS. Model performance was therefore evaluated qualitatively by examining whether observed occurrence records align with plausible temperature gradients (Figure 3) and by assessing sampling effort through temporal histograms. While this approach effectively illustrates spatial patterns of climate exposure, it does not account for precipitation, land-use change, habitat structure, or biotic interactions. Consequently, the results should be interpreted as an assessment of potential climatic exposure rather than a definitive prediction of future species distribution.

What I Learned

Through this project, I learned how to integrate biodiversity occurrence data with large-scale climate datasets in R, how to process and visualize raster-based climate projections, and how to compare present and future scenarios using CMIP6 SSP pathways. I also gained experience in critically evaluating the scope and limitations of climate-based analyses and in clearly communicating results and uncertainties in an ecological context.